

Integration By Parts:

If we want to evaluate the integral $\int f(x)g'(x)dx$, we may use integration by parts

Formula:

$$\int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx + c$$

Results:

- $\int e^{ax} [af(x) + f'(x)] dx = e^{ax} f(x) + c$
- $\int u dv = uv - \int v du$

EXERCISE 3.4

Q.1 Evaluate the following integrals by parts.

(i) $\int x \sin x dx$

Solution:

$$\begin{aligned} & \int x \sin x dx \\ & \text{Integrating by parts} \\ & = x(-\cos x) - \int (-\cos x).1 dx \\ & = -x \cos x + \int \cos x dx \\ & = -x \cos x + \sin x + c \end{aligned}$$

(ii) $\int \ln x dx$

Solution:

$$\begin{aligned} & \int \ln x.1 dx \\ & \text{Integrating by parts} \\ & = (\ln x)(x) - \int x.\frac{1}{x} dx \\ & = x \ln x - \int 1 dx \\ & = x \ln x - x + c \end{aligned}$$

(iii) $\int x \ln x dx$

Solution:

$$\begin{aligned} & \int \ln x.x dx \\ & \text{Integrating by parts} \\ & = \ln x \left(\frac{x^2}{2} \right) - \int \left(\frac{x^2}{2} \right) \cdot \frac{1}{x} dx \end{aligned}$$

$$\begin{aligned} & = \frac{x^2}{2} \ln x - \frac{1}{2} \int x dx \\ & = \frac{x^2}{2} \ln x - \frac{1}{2} \left(\frac{x^2}{2} \right) + c \\ & = \frac{x^2}{2} \ln x - \frac{x^2}{4} + c \end{aligned}$$

(iv) $\int x^2 \ln x dx$

Solution:

$$\begin{aligned} & \int \ln x.x^2 dx \\ & \text{Integrating by parts} \\ & = \ln x \left(\frac{x^3}{3} \right) - \int \left(\frac{x^3}{3} \right) \cdot \frac{1}{x} dx \\ & = \frac{x^3}{3} \ln x - \frac{1}{3} \int x^2 dx \\ & = \frac{x^3}{3} \ln x - \frac{1}{3} \left(\frac{x^3}{3} \right) + c = \frac{x^3}{3} \ln x - \frac{x^3}{9} + c \end{aligned}$$

(v) $\int x^3 \ln x dx$

Solution:

$$\begin{aligned} & \int \ln x.x^3 dx \\ & \text{Integrating by parts} \\ & = \ln x \left(\frac{x^4}{4} \right) - \int \frac{x^4}{4} \cdot \frac{1}{x} dx \\ & = \frac{x^4}{4} (\ln x) - \frac{1}{4} \int x^3 dx \end{aligned}$$

$$= \frac{x^4}{4}(\ln x) - \frac{1}{4}\left(\frac{x^4}{4}\right) + c$$

$$= \frac{x^4}{4}(\ln x) - \frac{1}{16}x^4 + c$$

$$(vi) \quad \int x^4 \ln x \, dx$$

Solution:

$$\int \ln x \cdot x^4 \, dx$$

Integrating by parts

$$= \ln x \left(\frac{x^5}{5} \right) - \int \left(\frac{x^5}{5} \right) \left(\frac{1}{x} \right) dx$$

$$= \frac{x^5}{5}(\ln x) - \frac{1}{5} \int x^4 \, dx$$

$$= \frac{x^5}{5}(\ln x) - \frac{1}{5} \left(\frac{x^5}{5} \right) + c = \frac{x^5}{5} \ln x - \frac{x^5}{25} + c$$

$$(vii) \quad \int \tan^{-1} x \, dx$$

Solution:

$$\int \tan^{-1} x \cdot 1 \, dx$$

Integrating by parts

$$= \tan^{-1} x(x) - \int x \cdot \frac{1}{1+x^2} \, dx$$

$$= x \tan^{-1} x - \frac{1}{2} \int \frac{2x}{1+x^2} \, dx$$

$$= x \tan^{-1} x - \frac{1}{2} \ln|1+x^2| + c$$

$$(viii) \quad \int x^2 \sin x \, dx$$

Solution:

$$\int x^2 \sin x \, dx$$

Integrating by parts

$$= x^2(-\cos x) - \int (-\cos x)(2x) \, dx$$

$$= -x^2 \cos x + 2 \int x \cos x \, dx$$

Integrating by parts

$$= -x^2 \cos x + 2 \left[x(\sin x) - \int \sin x \cdot 1 \, dx \right]$$

$$= -x^2 \cos x + 2x \sin x - 2(-\cos x) + c$$

$$= -x^2 \cos x + 2x \sin x + 2 \cos x + c$$

$$(ix) \quad \int x^2 \tan^{-1} x$$

Solution:

$$\int \tan^{-1} x \cdot x^2 \, dx$$

Integrating by parts

$$= \tan^{-1} x \left(\frac{x^3}{3} \right) - \int \frac{x^3}{3} \cdot \frac{1}{1+x^2} \, dx$$

$$= \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \int \frac{x^3}{1+x^2} \, dx \dots\dots (i)$$

$\frac{x^3}{1+x^2}$ is an improper fraction,

therefore by long division

$$x^2 + 1 \overline{\begin{array}{r} x^3 \\ \underline{\pm x^3 \pm x} \\ -x \end{array}}$$

$$\text{So } \frac{x^3}{1+x^2} = x - \frac{x}{x^2+1}$$

Hence (i) becomes

$$= \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \left[\int \left(x - \frac{x}{x^2+1} \right) dx \right]$$

$$= \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \left[\int x \, dx - \frac{1}{2} \int \frac{2x}{x^2+1} \, dx \right]$$

$$= \frac{x^3}{3} \tan^{-1} x - \frac{1}{3} \left[\frac{x^2}{2} - \frac{1}{2} \ln(x^2+1) \right] + c$$

$$= \frac{x^3}{3} \tan^{-1} x - \frac{1}{6} x^2 + \frac{1}{6} \ln(x^2+1) + c$$

$$(x) \quad \int x \tan^{-1} x \, dx$$

Solution:

$$\int \tan^{-1} x \cdot x \, dx$$

Integrating by parts

$$= \tan^{-1} x \left(\frac{x^2}{2} \right) - \int \frac{x^2}{2} \left(\frac{1}{1+x^2} \right) dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} \, dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \left[\int \frac{(1+x^2-1)}{1+x^2} \, dx \right]$$

$$\begin{aligned}
&= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \left[\int \frac{1+x^2}{1+x^2} dx - \int \frac{1}{1+x^2} dx \right] \\
&= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \left[\int dx - \int \frac{1}{1+x^2} dx \right] \\
&= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \left[x - \tan^{-1} x \right] + c \\
&= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} x + \frac{1}{2} \tan^{-1} x + c \\
&= \left(\frac{1}{2} \tan^{-1} x \right) (x^2 + 1) - \frac{1}{2} x + c
\end{aligned}$$

(xi) $\int x^3 \tan^{-1} x \, dx$

Solution:

$$\int \tan^{-1} x \cdot x^3 \, dx$$

Integrating by parts

$$= \tan^{-1} x \left(\frac{x^4}{4} \right) - \int \left(\frac{x^4}{4} \cdot \frac{1}{1+x^2} \right) dx$$

$$= \frac{x^4}{4} \tan^{-1} x - \frac{1}{4} \int \frac{x^4}{1+x^2} dx \dots\dots (i)$$

$\frac{x^4}{1+x^2}$ is an improper fraction,

therefore by long division

$$\begin{array}{r}
x^2 + 1 \overline{) \begin{array}{c} x^4 \\ \underline{\pm x^4 \pm x^2} \\ -x^2 \\ \underline{\mp x^2 \mp 1} \\ 1 \end{array}}
\end{array}$$

$$\therefore \frac{x^4}{1+x^2} = (x^2 - 1) + \frac{1}{x^2 + 1}$$

So, (i) becomes

$$= \frac{x^4}{4} \tan^{-1} x - \frac{1}{4} \left[\int \left(x^2 - 1 + \frac{1}{1+x^2} \right) dx \right]$$

$$= \frac{x^4}{4} \tan^{-1} x - \frac{1}{4} \left[\int x^2 dx - \int dx + \int \frac{1}{1+x^2} dx \right]$$

$$= \frac{x^4}{4} \tan^{-1} x - \frac{1}{4} \left[\frac{x^3}{3} - x + \tan^{-1} x \right] + c$$

$$= \frac{x^4}{4} \tan^{-1} x - \frac{x^3}{12} + \frac{x}{4} - \frac{1}{4} \tan^{-1} x + c$$

$$= \frac{1}{4} (x^4 - 1) \tan^{-1} x - \frac{x^3}{12} + \frac{1}{4} x + c$$

(xii) $\int x^3 \cos x \, dx$

Solution:

$$\int x^3 \cos x \, dx$$

Integrating by parts

$$= x^3 (\sin x) - \int \sin x (3x^2) dx$$

$$= x^3 \sin x - 3 \int x^2 \sin x \, dx$$

Integrating by parts

$$= x^3 \sin x - 3 \left[x^2 (-\cos x) - \int (-\cos x) 2x dx \right]$$

$$= x^3 \sin x + 3x^2 \cos x - 6 \int x \cos x \, dx$$

Integrating by part

$$= x^3 \sin x + 3x^2 \cos x - 6 \left[x (\sin x) - \int (\sin x) \cdot 1 dx \right]$$

$$= x^3 \sin x + 3x^2 \cos x - 6x \sin x + 6 \cos x + c$$

(xiii) $\int \sin^{-1} x \, dx$

Solution:

$$\int \sin^{-1} x \cdot 1 \, dx$$

Integrating by parts

$$= \sin^{-1} x (x) - \int x \cdot \frac{1}{\sqrt{1-x^2}} dx$$

$$= x \sin^{-1} x - \left(\frac{-1}{2} \right) \int \frac{-2x}{\sqrt{1-x^2}} dx$$

$$= x \sin^{-1} x + \frac{1}{2} \int (1-x^2)^{-\frac{1}{2}} (-2x) dx$$

$$= x \sin^{-1} x + \frac{1}{2} \frac{(1-x^2)^{\frac{1}{2}}}{\left(\frac{1}{2} \right)} + c$$

$$= x \sin^{-1} x + \sqrt{1-x^2} + c$$

(xiv) $\int x \sin^{-1} x \, dx$

Solution:

$$\int \sin^{-1} x \cdot x \, dx$$

Integrating by parts

$$\begin{aligned}
&= \sin^{-1} x \left(\frac{x^2}{2} \right) - \int \frac{x^2}{2} \cdot \frac{1}{\sqrt{1-x^2}} dx \\
&= \frac{x^2}{2} \sin^{-1} x + \frac{1}{2} \int \frac{-x^2}{\sqrt{1-x^2}} dx \\
&= \frac{x^2}{2} \sin^{-1} x + \frac{1}{2} \left[\int \frac{(1-x^2)-1}{\sqrt{1-x^2}} dx \right] \\
&= \frac{x^2}{2} \sin^{-1} x + \frac{1}{2} \left[\int \frac{(1-x^2)}{\sqrt{1-x^2}} dx - \int \frac{1}{\sqrt{1-x^2}} dx \right] \\
&= \frac{x^2}{2} \sin^{-1} x + \frac{1}{2} \left[\int \sqrt{1-x^2} dx - \int \frac{1}{\sqrt{1-x^2}} dx \right] \dots\dots (i)
\end{aligned}$$

$$\text{As } \int \sqrt{a^2 - x^2} dx = \frac{a^2}{2} \sin^{-1} \frac{x}{a} + \frac{x}{2} \sqrt{a^2 - x^2} + c$$

$$\text{and } \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + c$$

So (i) becomes

$$= \frac{x^2}{2} \sin^{-1} x + \frac{1}{2} \left[\frac{1}{2} \sin^{-1} x + \frac{x}{2} \sqrt{1-x^2} - \sin^{-1} x \right] + c$$

$$= \frac{x^2}{2} \sin^{-1} x - \frac{1}{4} \sin^{-1} x + \frac{x\sqrt{1-x^2}}{4} + c$$

$$(xv) \quad \int e^x \sin x \cos x dx$$

Solution:

$$\text{Let } I = \int e^x (\sin x \cos x) dx$$

Integrating by parts

$$\begin{aligned}
I &= e^x \left(\frac{\sin^2 x}{2} \right) - \int \left(\frac{\sin^2 x}{2} \right) \cdot e^x dx \\
&= \frac{e^x \sin^2 x}{2} - \frac{1}{2} \int \left(\frac{1 - \cos 2x}{2} \right) e^x dx \\
&= \frac{e^x \sin^2 x}{2} - \frac{1}{4} \int e^x dx + \frac{1}{4} \int e^x \cos 2x dx
\end{aligned}$$

Integrating by parts

$$\begin{aligned}
&= \frac{e^x \sin^2 x}{2} - \frac{1}{4} e^x + \frac{1}{4} \left[e^x \left(\frac{\sin 2x}{2} \right) - \int \frac{\sin 2x}{2} e^x dx \right] + c_1 \\
&= \frac{e^x \sin^2 x}{2} - \frac{1}{4} e^x + \frac{e^x \sin 2x}{8} - \frac{1}{4} \int \frac{2 \sin x \cos x}{2} e^x dx + c_1 \\
I &= e^x \frac{\sin^2 x}{2} - \frac{1}{4} e^x + \frac{e^x \sin 2x}{8} - \frac{1}{4} \int e^x (\sin x \cos x) dx + c_1
\end{aligned}$$

$$I = e^x \frac{\sin^2 x}{2} - \frac{1}{4} e^x + \frac{e^x \sin 2x}{8} - \frac{1}{4} I + c_1$$

$$I + \frac{1}{4} I = e^x \frac{\sin^2 x}{2} - \frac{1}{4} e^x + \frac{e^x \sin 2x}{8} + c_1$$

$$\frac{5}{4} I = e^x \frac{\sin^2 x}{2} - \frac{1}{4} e^x + \frac{e^x \sin 2x}{8} + c_1$$

$$I = \frac{2}{5} e^x \sin^2 x - \frac{1}{5} e^x + \frac{1}{10} e^x \sin 2x + \frac{4}{5} c_1$$

$$\text{Put } \frac{4}{5} c_1 = c$$

$$I = \frac{1}{5} e^x \left[2 \sin^2 x - 1 + \frac{1}{2} \sin 2x \right] + c$$

$$(xvi) \quad \int x \sin x \cos x dx$$

Solution:

$$\int x (\sin x \cos x) dx$$

Integrating by parts

$$\begin{aligned}
&= x \left(\frac{\sin^2 x}{2} \right) - \int \left(\frac{\sin^2 x}{2} \right) \cdot (1) dx \\
&= \frac{x}{2} \sin^2 x - \frac{1}{4} \int (2 \sin^2 x) dx \\
&= \frac{x}{2} \sin^2 x - \frac{1}{4} \int (1 - \cos 2x) dx \\
&= \frac{x}{2} \sin^2 x - \frac{1}{4} \int 1 dx + \frac{1}{4} \int \cos 2x dx \\
&= \frac{x}{2} \sin^2 x - \frac{1}{4} x + \frac{1}{4} \frac{\sin 2x}{2} + c \\
&= \frac{x}{2} \sin^2 x - \frac{1}{4} x + \frac{1}{8} \sin 2x + c
\end{aligned}$$

$$(xvii) \quad \int x \cos^2 x dx$$

Solution:

$$\begin{aligned}
&\int x \cos^2 x dx \\
&= \int x \frac{(1 + \cos 2x)}{2} dx \because \cos^2 x = \frac{1 + \cos 2x}{2} \\
&= \frac{1}{2} \int x dx + \frac{1}{2} \int x \cos 2x dx
\end{aligned}$$

Integrating by parts

$$\begin{aligned}
&= \frac{1}{2} \left(\frac{x^2}{2} \right) + \frac{1}{2} \left[x \frac{\sin 2x}{2} - \int \frac{\sin 2x}{2} (1) dx \right] \\
&= \frac{1}{4} x^2 + \frac{x \sin 2x}{4} - \frac{1}{4} \int \sin 2x dx \\
&= \frac{1}{4} x^2 + \frac{x \sin 2x}{4} - \frac{1}{4} \left(\frac{-\cos 2x}{2} \right) + c \\
&= \frac{1}{4} x^2 + \frac{x \sin 2x}{4} + \frac{\cos 2x}{8} + c \\
&= \frac{1}{4} \left[x^2 + x \sin 2x + \frac{1}{2} \cos 2x \right] + c
\end{aligned}$$

(xviii) $\int x \sin^2 x dx$

Solution:

$$\begin{aligned}
&\int x \sin^2 x dx \\
&= \int x \left(\frac{1 - \cos 2x}{2} \right) dx \because \sin^2 x = \frac{1 - \cos 2x}{2} \\
&= \frac{1}{2} \int x dx - \frac{1}{2} \int x \cos 2x dx \\
&= \frac{1}{2} \left(\frac{x^2}{2} \right) - \frac{1}{2} \int x \cos 2x dx \\
&\text{Integrating by parts} \\
&= \frac{x^2}{4} - \frac{1}{2} \left[x \frac{\sin 2x}{2} - \int \frac{\sin 2x}{2} (1) dx \right] \\
&= \frac{x^2}{4} - \frac{1}{4} x \sin 2x + \frac{1}{4} \int \sin 2x dx \\
&= \frac{1}{4} \left[x^2 - x \sin 2x + \left(\frac{-\cos 2x}{2} \right) \right] + c \\
&= \frac{1}{4} \left[x^2 - x \sin 2x - \frac{1}{2} \cos 2x \right] + c
\end{aligned}$$

(xix) $\int (\ln x)^2 dx$

Solution:

$$\begin{aligned}
&\int (\ln x)^2 (1) dx \\
&\text{Integrating by parts} \\
&= (\ln x)^2 (x) - \int x \cdot 2(\ln x) \cdot \frac{1}{x} dx \\
&= x(\ln x)^2 - 2 \int \ln x \cdot 1 dx \\
&\text{Integrating by parts}
\end{aligned}$$

$$\begin{aligned}
&= x(\ln x)^2 - 2 \left[(\ln x) \cdot x - \int x \cdot \frac{1}{x} dx \right] \\
&= x(\ln x)^2 - 2x \ln x + 2 \int 1 dx \\
&= x(\ln x)^2 - 2x \ln x + 2x + c \\
&= x \ln x [\ln x - 2] + 2x + c
\end{aligned}$$

(xx) $\int \ln(\tan x) \sec^2 x dx$

Solution:

$$\begin{aligned}
&\int \ln(\tan x) \sec^2 x dx \\
&\text{Integrating by parts} \\
&= \ln(\tan x) (\tan x) - \int (\tan x) \cdot \frac{1}{\tan x} \sec^2 x dx \\
&= \tan x \ln(\tan x) - \int \sec^2 x dx \\
&= \tan x \ln(\tan x) - \tan x + c
\end{aligned}$$

(xxi) $\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx$

Solution:

$$\begin{aligned}
&\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx \\
&= \int \sin^{-1} x \cdot \frac{x}{\sqrt{1-x^2}} dx \\
&= -\frac{1}{2} \left[\int \sin^{-1} x \left\{ (1-x^2)^{-\frac{1}{2}} (-2x) \right\} dx \right] \\
&\text{Integrating by parts} \\
&= -\frac{1}{2} \left[\sin^{-1} x \cdot \frac{\sqrt{1-x^2}}{\left(\frac{1}{2} \right)} - \int \frac{\sqrt{1-x^2}}{\left(\frac{1}{2} \right)} \cdot \frac{1}{\sqrt{1-x^2}} dx \right] \\
&= -\sqrt{1-x^2} \sin^{-1} x + \int 1 dx \\
&= -\sqrt{1-x^2} \sin^{-1} x + x + c
\end{aligned}$$

Q.2 Evaluate the following Integrals.

(i) $\int \tan^4 x dx$

Solution:

$$\begin{aligned}
&\int \tan^4 x dx \\
&= \int \tan^2 x \cdot \tan^2 x dx \\
&= \int \tan^2 x (\sec^2 x - 1) dx
\end{aligned}$$

$$\begin{aligned}
&= \int (\tan^2 x \sec^2 x - \tan^2 x) dx \\
&= \int (\tan x)^2 \sec^2 x dx - \int \tan^2 x dx \\
&= \frac{\tan^3 x}{3} - \int (\sec^2 x - 1) dx \\
&= \frac{\tan^3 x}{3} - \int \sec^2 x dx + \int 1 dx \\
&= \frac{\tan^3 x}{3} - \tan x + x + c
\end{aligned}$$

(ii) $\int \sec^4 x dx$

Solution:

$$\begin{aligned}
&\int \sec^4 x dx \\
&= \int \sec^2 x \cdot \sec^2 x dx \\
&= \int (1 + \tan^2 x) \sec^2 x dx \\
&= \int \sec^2 x dx + \int \tan^2 x \sec^2 x dx \\
&= \tan x + \frac{\tan^3 x}{3} + c
\end{aligned}$$

(iii) $\int e^x \sin 2x \cos x dx$

Solution:

$$\begin{aligned}
\text{Let } I &= \int e^x \sin 2x \cos x dx \\
I &= \frac{1}{2} \int [e^x 2 \sin 2x \cos x] dx \\
\text{Using } 2 \sin \alpha \cos \beta &= \sin(\alpha + \beta) + \sin(\alpha - \beta) \\
I &= \frac{1}{2} \int e^x [\sin(2x + x) + \sin(2x - x)] dx \\
I &= \frac{1}{2} \int e^x (\sin 3x + \sin x) dx \\
I &= \frac{1}{2} \int e^x \sin 3x dx + \frac{1}{2} \int e^x \sin x dx \\
&= \frac{1}{2} I_1 + \frac{1}{2} I_2 \dots (i)
\end{aligned}$$

Now $I_1 = \int e^x \sin 3x dx$

Integrating by parts

$$I_1 = e^x \left(\frac{-\cos 3x}{3} \right) - \int \left(\frac{-\cos 3x}{3} \right) \cdot e^x dx$$

$$I_1 = \frac{-\cos 3x}{3} e^x + \frac{1}{3} \int e^x \cos 3x dx$$

Integrating by parts

$$I_1 = \frac{-\cos 3x e^x}{3} + \frac{1}{3} \left[e^x \frac{\sin 3x}{3} - \int \frac{\sin 3x}{3} \cdot e^x dx \right]$$

$$I_1 = \frac{-\cos 3x}{3} e^x + \frac{1}{9} e^x \sin 3x - \frac{1}{9} \int e^x \sin 3x dx$$

$$I_1 = \frac{-\cos 3x}{3} e^x + \frac{1}{9} e^x \sin 3x - \frac{1}{9} I_1 + c_1$$

$$I_1 + \frac{1}{9} I_1 = \frac{-\cos 3x}{3} e^x + \frac{1}{9} e^x \sin 3x + c_1$$

$$\frac{10}{9} I_1 = \frac{-\cos 3x}{3} e^x + \frac{1}{9} e^x \sin 3x + c_1$$

$$I_1 = \frac{9}{10} \left[\frac{-\cos 3x}{3} e^x + \frac{\sin 3x}{9} e^x \right] + \frac{9}{10} c_1$$

$$I_1 = \frac{e^x}{10} [-3 \cos 3x + \sin 3x] + \frac{9}{10} c_1$$

Now

$$I_2 = \int e^x \sin x dx$$

Integrating by parts

$$I_2 = e^x (-\cos x) - \int (-\cos x) \cdot e^x dx$$

$$I_2 = -e^x \cos x + \int e^x \cos x dx$$

Integrating by parts

$$I_2 = -e^x \cos x + \left[e^x \sin x - \int \sin x e^x dx \right]$$

$$I_2 = -e^x \cos x + e^x \sin x - I_2 + c_2$$

$$2I_2 = -e^x \cos x + e^x \sin x + c_2$$

$$I_2 = \frac{1}{2} e^x [\sin x - \cos x] + \frac{1}{2} c_2$$

By putting the values of I_1 and I_2 in

$$(i) I = \frac{1}{2} \left[\frac{e^x}{10} (\sin 3x - 3 \cos 3x) + \frac{9}{10} c_1 \right]$$

$$+\frac{1}{2}\left[\frac{1}{2}e^x(\sin x - \cos x) + \frac{1}{2}c_2\right]$$

$$I = \frac{1}{2}e^x\left[\frac{1}{10}\sin 3x - \frac{3}{10}\cos 3x + \frac{1}{2}\sin x - \frac{1}{2}\cos x\right] + \frac{9}{20}c_1 + \frac{1}{4}c_2$$

$$I = \frac{1}{4}e^x\left[\frac{1}{5}\sin 3x - \frac{3}{5}\cos 3x + \sin x - \cos x\right] + c$$

$$\text{Where } c = \frac{9}{20}c_1 + \frac{1}{4}c_2$$

(iv) $\int \tan^3 x \sec x \, dx$

Solution:

$$\text{Let } I = \int \tan^3 x \sec x \, dx$$

$$I = \int \tan^2 x (\tan x \sec x) \, dx$$

Integrating by parts

$$I = \tan^2 x (\sec x) - \int \sec x (2 \tan x \sec^2 x) \, dx$$

$$I = \sec x \tan^2 x - 2 \int \sec x \tan x (1 + \tan^2 x) \, dx$$

$$I = \sec x \tan^2 x - 2 \int \sec x \tan x \, dx - 2 \int \tan^3 x \sec x \, dx$$

$$I = \sec x \tan^2 x - 2 \sec x - 2I + c_1$$

$$3I = \sec x \tan^2 x - 2 \sec x + c$$

$$I = \frac{1}{3} \left[\sec x \tan^2 x - 2 \sec x \right] + \frac{c_1}{3}$$

$$I = \frac{1}{3} \left[\sec x \tan^2 x - 2 \sec x \right] + c$$

$$\text{Where } c = \frac{c_1}{3}$$

(v) $\int x^3 e^{5x} \, dx$

Solution:

$$\int x^3 e^{5x} \, dx$$

Integrating by parts

$$= x^3 \left(\frac{e^{5x}}{5} \right) - \int \frac{e^{5x}}{5} (3x^2) \, dx$$

Integrating by parts

$$= \frac{x^3 e^{5x}}{5} - \frac{3}{5} \left[x^2 \left(\frac{e^{5x}}{5} \right) - \int \frac{e^{5x}}{5} (2x) \, dx \right]$$

$$= \frac{x^3 e^{5x}}{5} - \frac{3x^2 e^{5x}}{25} + \frac{6}{25} \int x e^{5x} \, dx$$

Integrating by parts

$$= \frac{x^3 e^{5x}}{5} - \frac{3x^2 e^{5x}}{25} + \frac{6}{25} \left[x \cdot \frac{e^{5x}}{5} - \int \frac{e^{5x}}{5} \cdot 1 \, dx \right]$$

$$= \frac{x^3 e^{5x}}{5} - \frac{3x^2 e^{5x}}{25} + \frac{6x e^{5x}}{125} - \frac{6}{125} \int e^{5x} \, dx$$

$$= \frac{x^3 e^{5x}}{5} - \frac{3x^2 e^{5x}}{25} + \frac{6x e^{5x}}{125} - \frac{6}{125} \left(\frac{e^{5x}}{5} \right) + c$$

$$= \frac{x^3 e^{5x}}{5} - \frac{3x^2 e^{5x}}{25} + \frac{6x e^{5x}}{125} - \frac{6e^{5x}}{625} + c$$

$$= \frac{1}{5} e^{5x} \left[x^3 - \frac{3x^2}{5} + \frac{6}{25} x - \frac{6}{125} \right] + c$$

(vi) $\int e^{-x} \sin 2x \, dx$

Solution:

$$\text{Let } I = \int e^{-x} \sin 2x \, dx$$

Integrating by parts

$$I = e^{-x} \left(\frac{-\cos 2x}{2} \right) - \int \left(\frac{-\cos 2x}{2} \right) (-1) e^{-x} \, dx$$

$$I = \frac{-e^{-x}}{2} \cos 2x - \frac{1}{2} \int \cos 2x \cdot e^{-x} \, dx$$

Integrating by parts

$$I = \frac{-e^{-x}}{2} \cos 2x - \frac{1}{2} e^{-x} \left[\left(\frac{\sin 2x}{2} \right) - \int \frac{\sin 2x}{2} (e^{-x}) (-1) \, dx \right]$$

$$I = \frac{-e^{-x}}{2} \cos 2x - \frac{1}{4} e^{-x} \sin 2x - \frac{1}{4} \int e^{-x} \sin 2x \, dx + c_1$$

$$I = \frac{-e^{-x}}{2} \cos 2x - \frac{1}{4} e^{-x} \sin 2x - \frac{1}{4} I + c_1$$

$$I + \frac{1}{4} I = \frac{-e^{-x} \cos 2x}{2} - \frac{1}{4} e^{-x} \sin 2x + c_1$$

$$\left(\frac{5}{4} \right) I = \frac{-e^{-x}}{2} \cos 2x - \frac{1}{4} e^{-x} \sin 2x + c_1$$

$$I = \frac{4}{5} \left(\frac{-e^{-x}}{2} \cos 2x - \frac{1}{4} e^{-x} \sin 2x \right) + \frac{4}{5} c_1$$

$$I = \frac{-2}{5} e^{-x} \left[\cos 2x + \frac{1}{2} \sin 2x \right] + c$$

$$\text{Where } \frac{4}{5} c_1 = c$$

(vii) $\int e^{2x} \cos 3x \, dx$

Solution:

$$\text{Let } I = \int e^{2x} \cos 3x dx$$

Integrating by parts

$$I = e^{2x} \frac{\sin 3x}{3} - \int \frac{\sin 3x}{3} e^{2x} \times 2 dx$$

$$I = \frac{\sin 3x}{3} e^{2x} - \frac{2}{3} \int \sin 3x e^{2x} dx$$

Integrating by parts

$$I = \frac{\sin 3x}{3} e^{2x} - \frac{2}{3} \left[e^{2x} \left(\frac{-\cos 3x}{3} \right) - \int \frac{-\cos 3x}{3} e^{2x} (2) dx + c_1 \right]$$

$$I = \frac{\sin 3x}{3} e^{2x} + \frac{2}{9} e^{2x} \cos 3x - \frac{4}{9} \int e^{2x} \cos 3x dx + c_1$$

$$I = \frac{\sin 3x}{3} e^{2x} + \frac{2}{9} e^{2x} \cos 3x - \frac{4}{9} I + c_1$$

$$I + \frac{4}{9} I = \frac{\sin 3x}{3} e^{2x} + \frac{2}{9} e^{2x} \cos 3x + c_1$$

$$\frac{13}{9} I = \frac{\sin 3x}{3} e^{2x} + \frac{2}{9} e^{2x} \cos 3x + c_1$$

$$I = \frac{9}{13} \left[\frac{\sin 3x}{3} e^{2x} + \frac{2}{9} e^{2x} \cos 3x \right] + \frac{9}{13} c_1$$

$$I = \frac{e^{2x}}{13} [3 \sin 3x + 2 \cos 3x] + c$$

$$\text{Where } c = \frac{9}{13} c_1$$

(viii) $\int \operatorname{cosec}^3 x dx$

Solution:

$$\text{Let } I = \int \operatorname{cosec}^3 x dx$$

$$I = \int \operatorname{cosec} x (\operatorname{cosec}^2 x) dx$$

Integrating by parts

$$I = \operatorname{cosec} x (-\cot x) - \int (-\cot x) (-\operatorname{cosec} x \cot x) dx$$

$$I = -\cot x \operatorname{cosec} x - \int \cot^2 x \operatorname{cosec} x dx$$

$$I = -\cot x \operatorname{cosec} x - \int (\operatorname{cosec}^2 x - 1) \operatorname{cosec} x dx$$

$$I = -\cot x \operatorname{cosec} x - \int \operatorname{cosec}^3 x dx + \int \operatorname{cosec} x dx$$

$$I = -\cot x \operatorname{cosec} x - I + \ln |\operatorname{cosec} x - \cot x| + c_1$$

$$2I = -\cot x \operatorname{cosec} x + \ln |\operatorname{cosec} x - \cot x| + c_1$$

$$I = -\frac{1}{2} [\cot x \operatorname{cosec} x - \ln |\operatorname{cosec} x - \cot x|] + \frac{1}{2} c_1$$

$$I = -\frac{1}{2} [\cot x \operatorname{cosec} x - \ln |\operatorname{cosec} x - \cot x|] + c$$

$$\text{Where } c = \frac{1}{2} c_1$$

Q.3 Show that

$$\int e^{ax} \sin bx dx = \frac{1}{\sqrt{a^2 + b^2}} e^{ax} \sin \left(bx - \tan^{-1} \frac{b}{a} \right) + c$$

Solution:

$$\text{Let } I = \int e^{ax} \sin bx dx$$

Integrating by parts

$$I = e^{ax} \left(-\frac{\cos bx}{b} \right) - \int \left(-\frac{\cos bx}{b} \right) (e^{ax}) (a) dx$$

$$I = \frac{-e^{ax}}{b} \cos bx + \frac{a}{b} \int e^{ax} \cos bx dx$$

Integrating by parts

$$I = \frac{-e^{ax}}{b} \cos bx + \frac{a}{b} \left[e^{ax} \left(\frac{\sin bx}{b} \right) - \int \frac{\sin bx}{b} (e^{ax}) (a) dx \right]$$

$$I = \frac{-e^{ax}}{b} \cos bx + \frac{a}{b} \left[\frac{e^{ax}}{b} \sin bx - \frac{a}{b} \int e^{ax} \sin bx dx \right]$$

$$I = \frac{-e^{ax}}{b} \cos bx + \frac{a}{b^2} e^{ax} \sin bx - \frac{a^2}{b^2} \int e^{ax} \sin bx dx + c_1$$

$$I = \frac{-e^{ax}}{b} \cos bx + \frac{a}{b^2} e^{ax} \sin bx - \frac{a^2}{b^2} I + c_1$$

$$I + \frac{a^2}{b^2} I = \frac{-e^{ax}}{b} \cos bx + \frac{a}{b^2} e^{ax} \sin bx + c_1$$

$$\left(\frac{a^2 + b^2}{b^2} \right) I = e^{ax} \left[\frac{-b \cos bx + a \sin bx}{b^2} \right] + c$$

$$I = \left(\frac{b^2}{a^2 + b^2} \right) \frac{e^{ax}}{b^2} [-b \cos bx + a \sin bx] + c_1 \left(\frac{b^2}{a^2 + b^2} \right)$$

$$I = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx] + c$$

$$\text{Where } c = c_1 \left(\frac{b^2}{a^2 + b^2} \right)$$

$$\text{Put } a = r \cos \theta, b = r \sin \theta$$

$$a^2 + b^2 = r^2 (\cos^2 \theta + \sin^2 \theta) \text{ and } \tan \theta = \frac{b}{a}$$

$$\Rightarrow r = \sqrt{a^2 + b^2} \text{ and } \theta = \tan^{-1} \frac{b}{a}$$

$$I = \frac{1}{a^2 + b^2} e^{ax} [r \cos \theta \sin bx - r \sin \theta \cos bx] + c$$

$$I = \frac{r}{a^2 + b^2} e^{ax} [\cos \theta \sin bx - \sin \theta \cos bx] + c$$

$$\text{Using } \sin \alpha \cos \beta - \cos \alpha \sin \beta = \sin(\alpha - \beta)$$

$$I = \frac{\sqrt{a^2 + b^2}}{a^2 + b^2} e^{ax} \sin(bx - \theta) + c$$

$$I = \frac{1}{\sqrt{a^2 + b^2}} e^{ax} \sin\left(bx - \tan^{-1}\left(\frac{b}{a}\right)\right) + c$$

Q.4 Evaluate the following indefinite integrals.

(i) $\int \sqrt{a^2 - x^2} dx$

Solution:

$$\text{Let } I = \int \sqrt{a^2 - x^2} dx$$

$$I = \int \sqrt{a^2 - x^2} (1) dx$$

Integrating by parts

$$I = \sqrt{a^2 - x^2} (x) - \int x \cdot \frac{1}{2} (a^2 - x^2)^{-\frac{1}{2}} (-2x) dx$$

$$I = x\sqrt{a^2 - x^2} + \int \frac{x^2}{\sqrt{a^2 - x^2}} dx$$

$$I = x\sqrt{a^2 - x^2} + \int \frac{a^2 - a^2 + x^2}{\sqrt{a^2 - x^2}} dx$$

$$I = x\sqrt{a^2 - x^2} + \int \frac{a^2 - (a^2 - x^2)}{\sqrt{a^2 - x^2}} dx$$

$$I = x\sqrt{a^2 - x^2} + \int \frac{a^2}{\sqrt{a^2 - x^2}} dx - \int \frac{a^2 - x^2}{\sqrt{a^2 - x^2}} dx$$

$$I = x\sqrt{a^2 - x^2} + a^2 \int \frac{dx}{\sqrt{a^2 - x^2}} - \int \sqrt{a^2 - x^2} dx$$

$$I = x\sqrt{a^2 - x^2} + a^2 \sin^{-1}\left(\frac{x}{a}\right) - I + c_1$$

$$2I = x\sqrt{a^2 - x^2} + a^2 \sin^{-1}\left(\frac{x}{a}\right) + c_1$$

$$I = \frac{1}{2} \left[x\sqrt{a^2 - x^2} + a^2 \sin^{-1}\left(\frac{x}{a}\right) \right] + \frac{c_1}{2}$$

$$I = \frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + c$$

$$\text{Where } \frac{c_1}{2} = c$$

(ii) $\int \sqrt{x^2 - a^2} dx$

Solution:

$$\text{Let } I = \int \sqrt{x^2 - a^2} dx$$

$$I = \int \sqrt{x^2 - a^2} (1) dx$$

Integrating by parts

$$I = \sqrt{x^2 - a^2} (x) - \int x \cdot \frac{1}{2} (x^2 - a^2)^{-\frac{1}{2}} (2x) dx$$

$$I = x\sqrt{x^2 - a^2} - \int \frac{x^2}{\sqrt{x^2 - a^2}} dx$$

$$I = x\sqrt{x^2 - a^2} - \int \frac{(x^2 - a^2) + a^2}{\sqrt{x^2 - a^2}} dx$$

$$I = x\sqrt{x^2 - a^2} - \int \frac{x^2 - a^2}{\sqrt{x^2 - a^2}} dx - \int \frac{a^2}{\sqrt{x^2 - a^2}} dx$$

$$I = x\sqrt{x^2 - a^2} - \int \sqrt{x^2 - a^2} dx - a^2 \int \frac{1}{\sqrt{x^2 - a^2}} dx$$

$$\text{Using } \int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln |x + \sqrt{x^2 - a^2}| + c$$

$$I = x\sqrt{x^2 - a^2} - I - a^2 \ln |x + \sqrt{x^2 - a^2}| + c_1$$

$$2I = x\sqrt{x^2 - a^2} - a^2 \ln |x + \sqrt{x^2 - a^2}| + c_1$$

$$I = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln |x + \sqrt{x^2 - a^2}| + \frac{c_1}{2}$$

$$\text{Where } c = \frac{c_1}{2}$$

$$I = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln |x + \sqrt{x^2 - a^2}| + c$$

(iii) $\int \sqrt{4 - 5x^2} dx$

Solution:

$$\text{Let } I = \int \sqrt{4 - 5x^2} (1) dx$$

Integrating by parts

$$I = \sqrt{4 - 5x^2} (x) - \int x \cdot \frac{1}{2} (4 - 5x^2)^{-\frac{1}{2}} (-10x) dx$$

$$I = x\sqrt{4 - 5x^2} - \int \frac{-5x^2}{\sqrt{4 - 5x^2}} dx$$

$$I = x\sqrt{4-5x^2} - \int \frac{(4-5x^2)-4}{\sqrt{4-5x^2}} dx$$

$$I = x\sqrt{4-5x^2} - \int \frac{(4-5x^2)}{\sqrt{4-5x^2}} dx + 4 \int \frac{1}{\sqrt{4-5x^2}} dx$$

$$I = x\sqrt{4-5x^2} - \int \sqrt{4-5x^2} dx + 4 \int \frac{dx}{\sqrt{(2)^2 - (\sqrt{5}x)^2}}$$

$$I = x\sqrt{4-5x^2} - I + 4 \left(\frac{\sin^{-1}\left(\frac{\sqrt{5}x}{2}\right)}{\sqrt{5}} \right) + c_1$$

$$2I = x\sqrt{4-5x^2} + \frac{4}{\sqrt{5}} \sin^{-1} \frac{\sqrt{5}x}{2} + c_1$$

$$I = \frac{x}{2} \sqrt{4-5x^2} + \frac{2}{\sqrt{5}} \sin^{-1} \frac{\sqrt{5}x}{2} + c$$

Where $c = \frac{c_1}{2}$

(iv) $\int \sqrt{3-4x^2} dx$

Solution:

Let $I = \int \sqrt{3-4x^2} \cdot (1) dx$

Integrating by parts

$$I = \sqrt{3-4x^2} (x) - \int x \cdot \frac{1}{2} (3-4x^2)^{-\frac{1}{2}} (-8x) dx$$

$$I = x\sqrt{3-4x^2} - \int \frac{-4x^2}{\sqrt{3-4x^2}} dx$$

$$I = x\sqrt{3-4x^2} - \int \frac{(3-4x^2)-3}{\sqrt{3-4x^2}} dx$$

$$I = x\sqrt{3-4x^2} - \int \frac{3-4x^2}{\sqrt{3-4x^2}} dx + 3 \int \frac{1}{\sqrt{3-4x^2}} dx$$

$$I = x\sqrt{3-4x^2} - \int \sqrt{3-4x^2} dx + 3 \int \frac{1}{\sqrt{(\sqrt{3})^2 - (2x)^2}} dx$$

$$I = x\sqrt{3-4x^2} - I + 3 \left(\frac{1}{2} \sin^{-1} \left(\frac{2x}{\sqrt{3}} \right) \right) + c_1$$

$$2I = x\sqrt{3-4x^2} + \frac{3}{2} \sin^{-1} \left(\frac{2x}{\sqrt{3}} \right) + c_1$$

$$I = \frac{x}{2} \sqrt{3-4x^2} + \frac{3}{4} \sin^{-1} \frac{2x}{\sqrt{3}} + c$$

Where $c = \frac{c_1}{2}$

(v) $\int \sqrt{x^2+4} dx$

Solution:

Let $I = \int \sqrt{x^2+4} \cdot (1) dx$

Integrating by parts

$$I = x\sqrt{x^2+4} - \int x \cdot \frac{1}{2} (x^2+4)^{-\frac{1}{2}} (2x) dx$$

$$I = x\sqrt{x^2+4} - \int \frac{x^2}{\sqrt{x^2+4}} dx$$

$$I = x\sqrt{x^2+4} - \int \frac{(x^2+4)-4}{\sqrt{x^2+4}} dx$$

$$I = x\sqrt{x^2+4} - \int \frac{x^2+4}{\sqrt{x^2+4}} dx + \int \frac{4}{\sqrt{x^2+4}} dx$$

$$I = x\sqrt{x^2+4} - \int \sqrt{x^2+4} dx + 4 \int \frac{1}{\sqrt{x^2+4}} dx$$

$$I = x\sqrt{x^2+4} - \int \sqrt{x^2+4} dx + 4 \int \frac{1}{\sqrt{x^2+2^2}} dx$$

$$\therefore \int \frac{1}{\sqrt{x^2+2^2}} dx = \ln |x + \sqrt{x^2+4}| + c_1$$

$$I = x\sqrt{x^2+4} - I + 4 \ln |x + \sqrt{x^2+4}| + c_1$$

$$2I = x\sqrt{x^2+4} + 4 \ln |x + \sqrt{x^2+4}| + c_1$$

$$I = \frac{x}{2} \sqrt{x^2+4} + 2 \ln |x + \sqrt{x^2+4}| + c$$

Where $c = \frac{c_1}{2}$

(vi) $\int x^2 e^{ax} dx$

Solution:

$$\int x^2 e^{ax} dx$$

Integrating by parts

$$= x^2 \left(\frac{e^{ax}}{a} \right) - \int \left(\frac{e^{ax}}{a} \right) (2x) dx$$

$$\begin{aligned}
&= \frac{x^2}{a} e^{ax} - \frac{2}{a} \int x e^{ax} dx \\
&\text{Integrating by parts} \\
&= \frac{x^2}{a} e^{ax} - \frac{2}{a} \left[x \left(\frac{e^{ax}}{a} \right) - \int \left(\frac{e^{ax}}{a} \right) \cdot 1 dx \right] \\
&= \frac{x^2}{a} e^{ax} - \frac{2}{a} \left[\frac{x}{a} e^{ax} + \frac{1}{a} \int e^{ax} dx \right] \\
&= \frac{x^2}{a} e^{ax} - \frac{2}{a} \left[\frac{x}{a} e^{ax} + \frac{1}{a} \left(\frac{e^{ax}}{a} \right) \right] + c \\
&= \frac{x^2}{a} e^{ax} - \frac{2}{a^2} x e^{ax} + \frac{2}{a^3} e^{ax} + c \\
&= \frac{1}{a} e^{ax} \left[x^2 - \frac{2}{a} x + \frac{2}{a^2} \right] + c
\end{aligned}$$

Q.5 Evaluate the following indefinite integrals

(i) $\int e^x \left(\frac{1}{x} + \ln x \right) dx$

Solution:

$$\begin{aligned}
&\int e^x \left(\frac{1}{x} + \ln x \right) dx \\
&= \int e^x \left(\ln x + \frac{1}{x} \right) dx
\end{aligned}$$

$$\begin{aligned}
&\text{Using } \int e^{ax} (a f(x) + f'(x)) dx = e^{ax} f(x) + c \\
&= e^x \ln x + c
\end{aligned}$$

Alternate method:

$$\begin{aligned}
&\int e^x \left(\frac{1}{x} + \ln x \right) dx \\
&= \int e^x \cdot \frac{1}{x} dx + \int e^x \cdot \ln x dx
\end{aligned}$$

$$\begin{aligned}
&\text{Integrating by parts} \\
&= e^x \cdot \ln x - \int \ln x \cdot e^x dx + \int e^x \cdot \ln x dx \\
&= e^x \ln x + c
\end{aligned}$$

(ii) $\int e^x (\cos x + \sin x) dx$

Solution:

$$\int e^x (\cos x + \sin x) dx$$

$$= \int e^x (\sin x + \cos x) dx$$

$$\text{Using } \int e^{ax} (a f(x) + f'(x)) dx = e^{ax} f(x) + c$$

$$= e^x \sin x + c$$

(iii) $\int e^{ax} \left[a \sec^{-1} x + \frac{1}{x\sqrt{x^2-1}} \right] dx$

Solution:

$$\int e^{ax} \left[a \sec^{-1} x + \frac{1}{x\sqrt{x^2-1}} \right] dx$$

$$\text{Using } \int e^{ax} (a f(x) + f'(x)) dx = e^{ax} f(x) + c$$

$$= e^{ax} \sec^{-1} x + c$$

(iv) $\int e^{3x} \left(\frac{3 \sin x - \cos x}{\sin^2 x} \right) dx$

Solution:

$$\int e^{3x} \left(\frac{3 \sin x - \cos x}{\sin^2 x} \right) dx$$

$$= \int e^{3x} \left(\frac{3 \sin x}{\sin^2 x} - \frac{\cos x}{\sin^2 x} \right) dx$$

$$= \int e^{3x} [3 \operatorname{cosec} x - \operatorname{cosec} x \cot x] dx$$

$$= \int e^{3x} [3 \operatorname{cosec} x + (-\operatorname{cosec} x \cot x)] dx$$

$$\text{Using } \int e^{ax} (a f(x) + f'(x)) dx = e^{ax} f(x) + c$$

$$= e^{3x} \operatorname{cosec} x + c$$

(v) $\int e^{2x} (-\sin x + 2 \cos x) dx$

Solution:

$$\int e^{2x} (-\sin x + 2 \cos x) dx$$

$$= \int e^{2x} (2 \cos x + (-\sin x)) dx$$

$$\text{Using } \int e^{ax} (a f(x) + f'(x)) dx = e^{ax} f(x) + c$$

$$= e^{2x} \cos x + c$$

(vi) $\int \frac{x e^x}{(1+x)^2} dx$

Solution:

$$\begin{aligned}
 & \int \frac{xe^x}{(1+x)^2} \\
 &= \int \frac{(1+x-1)e^x}{(1+x)^2} dx \\
 &= \int e^x \left(\frac{(1+x)}{(1+x)^2} - \frac{1}{(1+x)^2} \right) dx \\
 &= \int e^x \left(\frac{1}{(1+x)} + \left(\frac{-1}{(1+x)^2} \right) \right) dx
 \end{aligned}$$

Using $\int e^{ax} (a f(x) + f'(x)) dx = e^{ax} f(x) + c$

$$= e^x \left(\frac{1}{1+x} \right) + c$$

(vii) $\int e^{-x} [\cos x - \sin x] dx$

Solution:

$$\begin{aligned}
 & \int e^{-x} [\cos x - \sin x] dx \\
 &= \int e^{-x} [-\sin x + \cos x] dx
 \end{aligned}$$

Using $\int e^{ax} (a f(x) + f'(x)) dx = e^{ax} f(x) + c$

$$= e^{-x} \sin x + c$$

(viii) $\int \frac{e^{m \tan^{-1} x}}{1+x^2} dx$

Solution:

$$\begin{aligned}
 & \int \frac{e^{m \tan^{-1} x}}{1+x^2} dx \\
 &= \int e^{m \tan^{-1} x} \frac{1}{1+x^2} dx
 \end{aligned}$$

Let $\tan^{-1} x = t$ then $\left(\frac{1}{1+x^2} \right) dx = dt$

$$= \int e^{mt} dt$$

$$= \frac{e^{mt}}{m} + c$$

$$= \frac{1}{m} e^{m \tan^{-1} x} + c$$

(ix) $\int \frac{2x}{1-\sin x} dx$

Solution:

$$\int \frac{2x}{1-\sin x} dx$$

Using $\cos\left(\frac{\pi}{2} - x\right) = \sin x$

$$= \int \frac{2x}{1-\cos\left(\frac{\pi}{2} - x\right)} dx$$

$$\because 1-\cos\left(\frac{\pi}{2} - x\right) = 2\sin^2\left(\frac{\pi}{4} - \frac{x}{2}\right)$$

$$= \int \frac{2x}{2\sin^2\left(\frac{\pi}{4} - \frac{x}{2}\right)} dx$$

$$= \int x \operatorname{cosec}^2\left(\frac{\pi}{4} - \frac{x}{2}\right) dx$$

Integrating by parts

$$= x(-2) \left(-\cot\left(\frac{\pi}{4} - \frac{x}{2}\right) \right) - \int (-2) \left(-\cot\left(\frac{\pi}{4} - \frac{x}{2}\right) \right) 1 dx$$

$$= 2x \cot\left(\frac{\pi}{4} - \frac{x}{2}\right) - 2 \int \cot\left(\frac{\pi}{4} - \frac{x}{2}\right) dx$$

$$= 2x \cot\left(\frac{\pi}{4} - \frac{x}{2}\right) - 2(-2) \ln \left| \sin\left(\frac{\pi}{4} - \frac{x}{2}\right) \right| + c$$

$$= 2x \cot\left(\frac{\pi}{4} - \frac{x}{2}\right) + 4 \ln \left| \sin\left(\frac{\pi}{4} - \frac{x}{2}\right) \right| + c$$

(x) $\int \frac{e^x (1+x)}{(2+x)^2} dx$

Solution:

$$\int \frac{e^x (1+x)}{(2+x)^2} dx$$

$$= \int e^x \frac{[2+x-1]}{(2+x)^2} dx$$

$$= \int e^x \left[\frac{2+x}{(2+x)^2} - \frac{1}{(2+x)^2} \right] dx$$

$$= \int e^x \left[\frac{1}{(2+x)} + \frac{-1}{(2+x)^2} \right] dx$$

$$\text{Using } \int e^{ax} (a f(x) + f'(x)) dx = e^{ax} f(x) + c$$

$$= e^x \left(\frac{1}{2+x} \right) + c$$

$$\text{(xi)} \quad \int \left(\frac{1-\sin x}{1-\cos x} \right) e^x dx$$

Solution:

$$\int \left(\frac{1-\sin x}{1-\cos x} \right) e^x dx$$

$$= \int \left(\frac{1-\sin x}{2 \sin^2 \frac{x}{2}} \right) e^x dx \because 1-\cos x = 2 \sin^2 \frac{x}{2}$$

$$= \int e^x \left[\frac{1}{2 \sin^2 \frac{x}{2}} - \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \sin^2 \frac{x}{2}} \right] dx \quad \because \sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$= \int e^x \left[\frac{1}{2} \operatorname{cosec}^2 \frac{x}{2} - \cot \frac{x}{2} \right] dx$$

$$= \int e^x \left[\left(-\cot \frac{x}{2} \right) + \left(\frac{1}{2} \operatorname{cosec}^2 \frac{x}{2} \right) \right] dx$$

$$\text{Using } \int e^{ax} (a f(x) + f'(x)) dx = e^{ax} f(x) + c$$

$$= e^x \left(-\cot \frac{x}{2} \right) + c \quad \because f(x) = -\cot \frac{x}{2}$$

$$= -e^x \cot \frac{x}{2} + c$$